Université Catholique d'Afrique Centrale (UCAC)

Institut Catholique de Yaoundé (ICY)

LEVEL THREE FIRST SEMESTER CA: 2025

DEPARTMENTS: MFEB
COURSE TITLE: OPERATION RESEARCH
DURATION: 02 HOURS
LECTURER: ARREY Marinus ARREY
INSTRUCTIONS: ANSWER ALL QUESTIONS

Question 1 (8marks)

The ABC Company has been a producer of picture tubes for television sets and certain printed circuits for radios. The company has just expanded into full scale production and marketing of AM and FM radios. It has built a new plant that can operate 48 hours per week. Production of an AM radio in the new plant will require 2 hours and production of an FM radio will require 3 hours. Each AM radio will contribute 40frs to profits while an FM radio will contribute 80frs to profits. The marketing department, after extensive research has determined that a maximum of 15AM radios and 10 FM radios can be sold each week.

(a) Formulate a linear programming model to determine the optimum production mix of AM and FM radio that will maximize profits.

(b) Solve this problem using the graphical method.

Question 2 (12 marks)

Solve the following linear programing problem using the big M method.

Max $Z = 12X_1+15X_2+9X_3$ — 0

St $8X_1+16X_2+12X_3 \leq 250$

$4X_1+8X_2+10X_3 \geq 80$

$7X_1+9X_2+8X_3 = 105$

$X_1, X_2, X_3 \geq 0$

Good Luck!

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